

Late Breaking Results: Circuit-Algorithm Co-design for Learnable Audio Analog Front-End

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Abstract This paper presents a circuit-algorithm co-design framework for learnable audio analog front-end (AFE) which includes an analog filterbank for feature extraction and a classifier based on Depthwise Separable Convolutional Neural Network (DSCNN). Instead of the traditional approach to design the analog filterbank and digital classifier separately, a learnable filterbank is proposed and its source-follower bandpass filter (SF-BPF) parameters are optimized together with the neural network classifier in a signal-to-noise ratio (SNR)-aware training process. A new system criterion function (L_{BPF}) is proposed to include classification loss and filter performance into the training process. The optimized audio AFE achieves 10.6% and 11.7% reduction in BPF power and chip area, respectively. Meanwhile, this approach achieved 88.6%–94.5% accuracy for 10-keyword classification task across a wide range of input signal SNR from 5dB to 20dB, with only 16k trainable parameters.

Index Terms Learnable filterbank, Analog bandpass filter, Design optimization, SNR-aware training, Keyword spotting

I. INTRODUCTION

Always on keyword spotting (KWS) is an emerging human-machine interface for mobile and Internet of Things (IoT) devices. Key performance metrics for KWS module include classification accuracy and power consumption. Feature extraction (FEx) is a pivotal function in KWS system. The use of analog FEx based on bandpass filterbank offers the advantage of higher energy efficiency compared to its digital counterpart, which requires analog-to-digital conversion [1]. However, analog FEx also faces the challenges such as channel overlapping, gain variation, and large on-chip capacitors [2]. Conventional filterbank design follows Mel frequency scale with uniform filter gain across all channels. However, achieving this result requires larger bias current to boost the transistor transconductance (g_m) for those high frequency channels.

Inspired by the recent work on learnable digital filter [3], this paper introduces a circuit-algorithm co-design framework for learnable audio analog front-end. The analog circuit parameters of a source follower BPF (SF BPF) filterbank are optimized together with the neural network classifier in a signal to noise ratio (SNR) aware training process for optimum overall system performance.

II. PROPOSED METHOD

The proposed audio AFE architecture for always-on KWS is illustrated in Fig. 1 (top). Voice signal from microphone goes through FEx to extract the channel energy and convert to digital features for classification. A depthwise separable CNN (DSCNN) network is used for 12 classes classification. Each FEx channel comprises a tunable SF BPF and a spectrogram generator. The schematic of the fully differential 2nd-order SF-BPF is shown in Fig. 1 (bottom). Transistors $M_{1,2}$ and $M_{3,4}$ serve as the super source followers, and floating capacitors $C_{1,2}$ determine the pole locations. Through small signal model

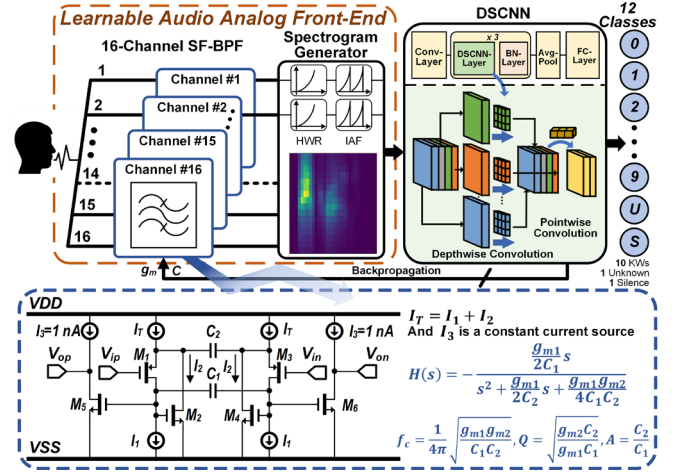


Fig. 1. The block diagram of the proposed learnable audio AFE and the schematic of the source-follower bandpass filter.

analysis, neglecting body effects, the filter transfer function, central frequency f_c , quality factor Q and passband gain A can be derived as shown in Fig. 1 (bottom right). Here, $g_{m1,2} = I_{1,2}/(nU_T)$, assuming the transistors work in subthreshold region ($n \approx 1.5$ as the subthreshold slope factor, and $U_T \approx 26$ mV as the thermal voltage at room temperature). It can be observed that the key performance of each BPF can be largely defined by 4 circuit parameters (g_{m1} , g_{m2} and C_1 , C_2). These parameters are included in the overall KWS system training process using back propagation. The initial circuit values are based on the reference design in SPICE circuit simulation. The trained values are feedback to update the circuit parameters. Instead of just achieving local optimums when optimizing the FEx front end and classification back end separately, the proposed approach seeks to achieve global optimum including both front end and back end.

The optimization boundaries such as f_c spans from 100 Hz to 5k Hz. It should be noted that some filter parameters such as g_m and C may have huge value difference up to the order of 10^3 or even more. Such differences complicate training due to its impact on the learning rates and potentially leading to vanishing gradient. To address this issue, two trainable scaling factors (φ_g and φ_C) are introduced in (1). These scaling factors represent the ratio of two coupled parameters, allowing for more balanced updates during the training process. Moreover, in this framework, C_1 is designated as the unit capacitor with a value of 3.2 pF, and g_{m1} is scaled to 3.84 nS. Both factors are scalable in real circuit design. This approach enables a flexible and realistic representation of capacitance values. With each channel's φ_g and φ_C , the SF BPF parameters can be written as

$$\varphi_g = \frac{g_{m2}}{g_{m1}}, \quad \varphi_C = \frac{C_2}{C_1} \quad (1)$$

$$f_c = \frac{1}{4\pi} \frac{g_{m1}}{C_1} \sqrt{\frac{\varphi_g}{\varphi_C}}, \quad Q = \sqrt{\varphi_g \varphi_C}, \quad A = \varphi_C \quad (2)$$

To optimize algorithm and circuit performance concurrently, we introduce a new loss function, L_{BPF} , as a more

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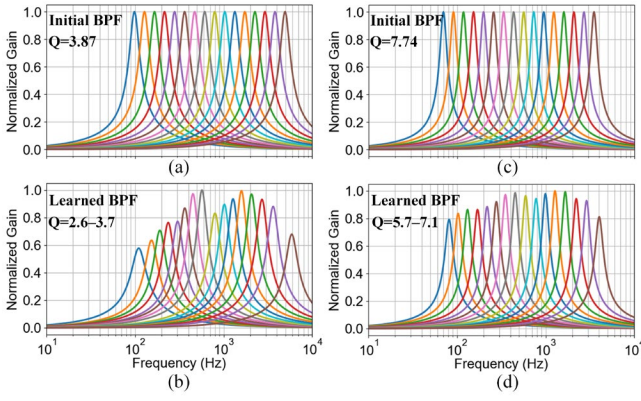


Fig. 2. Frequency response of (a) initial and (b) learned SF-BPF with an initial Q-factor 3.87. (c) and (d) showcase a version with high Q-factor.

effective system criterion for co-design optimization. In addition to the cross-entropy loss (L_{CE}) for KWS classification, power loss (L_P) and area loss (L_A) are incorporated into L_{BPF} to collectively minimize during training. As depicted in (3), L_P and L_A can be formulated in terms of φ_g and φ_c , respectively for all 16 channels. The power consumption of SF-BPF, $2V_{DD}(I_T + I_3)$, is linearly related to $g_{m1,2}$. Furthermore, capacitors contribute significantly to the area of the SF-BPF. Regularization coefficients λ_{CE} , λ_P and λ_A are introduced to adjust the relative importance of each loss term during model initialization.

$$L_{BPF} = L_{CE} + L_P + L_A$$

$$\rightarrow L_{BPF} = \lambda_{CE}L_{CE} + \lambda_P \sum_{i=1}^{16} \varphi_{g,i} + \lambda_A \sum_{i=1}^{16} \varphi_{c,i} \quad (3)$$

The elucidation of the equivalent circuit transfer function, adaptive scaling strategies, and the system criterion function collectively offer a comprehensive insight into the behaviour of the filter and the machine learning topology in joint optimization. The interplay between the circuitry's intricacies and the neural network algorithms enhances the understanding of the filter's behaviour, facilitating a sophisticated optimization strategy for the learnable audio front-end.

III. RESULTS AND DISCUSSION

The performance of the proposed design is evaluated with the Google Speech Command Dataset (GSCD). The initial configuration for g_{m1} , C_1 , and the scale of f_c between two consecutive channels, are based on a reference circuit designed in SPICE simulator. The frequency response, depicted in Fig. 2 (a), exhibits uniform gain and a Q-factor of 3.87 by initializing φ_g and φ_c to 2.0 and 7.5, respectively. Boundaries for φ_g and φ_c are imposed during training ($\varphi_g > 1$ and $\varphi_c > 1$) to ensure the optimization result adhere to circuit theory. Optimal SF-BPF frequency responses obtain via the training process in Fig. 2 (b) highlight nonuniform gains and Q-factors across the 16 channels. Moreover, in Fig. 3 (a), SNR-aware training enhances the noise resilience of SF-BPF, notably improving KWS accuracy, particularly at lower SNR, in contrast to the conventional noise-free training. The achieved classification accuracy reaches 87.8% and 94.2% under 5 dB and 20 dB SNR for a 12-class KWS task on GSCD with only 16k INT8 neural network parameters. Moreover, compared to directly training circuit component values (g_m and C), the learning of scaling factors (φ_g and φ_c) benefits both KWS performance and hardware utilization under different Q-factor levels. The comparison is shown in Fig. 3 (b), and a detailed summary is shown in Table I. The system performance is summarized and compared with other state-of-the-art designs in Table II, this design achieved the smallest model size with outstanding classification accuracy. This is

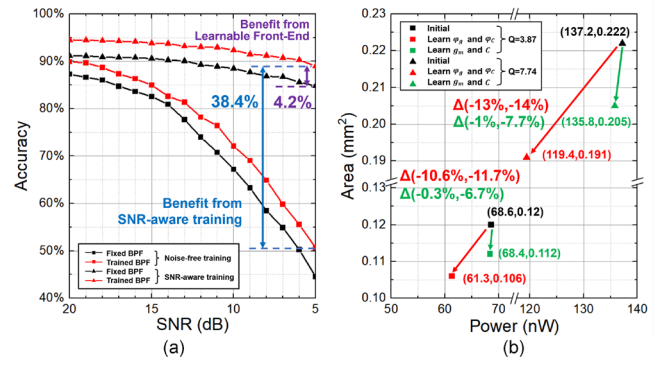


Fig. 3. (a) Noise-resilience through SNR-aware training. (b) Comparison of hardware utilization based on different learning strategies.

TABLE I. PERFORMANCE COMPARISON

	Low Quality BPF		High Quality BPF	
	Initial	Trained	Initial	Trained
φ_g	2.0	1.5 – 2.0	4.0	2.9 – 4.0
φ_c	7.5	4.5 – 7.8	15.0	11.0 – 13.9
Q	3.87	2.6 – 3.7	7.74	5.7 – 7.1
Power (nW)	68.6	61.3	137.2	119.4
Area* (mm ²)	0.120	0.106	0.222	0.191
Learning Method	Fix BPF	Train BPF and DSCNN	Fix BPF	Train BPF and DSCNN
Accuracy @ 5 dB SNR	84.7	88.6	86.3	89.0
Accuracy @ 20 dB SNR	91.1	94.5	92.6	94.6

*Based on post-layout simulation.

TABLE II. BENCHMARK TABLE

	Filter Type	Learnable Filter	Classifier	Trainable Parameters	Accuracy
VLSI' 19 [1]	Digital	No	LSTM	23k INT8	90.9%
ICLR' 20 [3]	Digital	Yes	CNN	4M FP32	93.4%
ICASSP' 23 [4]	Digital	Yes	ResNet	238k FP32	94.4%
VLSI' 22 [2]	Analog	No	ResNet	238k FP32	91.5%
This Work	Analog	Yes	DSCNN	16k INT8	94.5%

the first reported learnable analog front-end with circuit-algorithm co-design for audio acquisition.

IV. CONCLUSION

A learnable audio AFE with a DSCNN optimizes SF-BPF for always-on KWS is presented in this work. The incorporation of adaptive scaling strategies, a comprehensive equivalent circuit analysis, and a novel co-design optimization criterion function showcases a holistic framework for achieving superior KWS performance and efficient hardware resource utilization. The presented architecture achieves notable reductions of 10.6% and 11.7% in post-layout BPF power and area consumption, respectively, while maintaining outstanding classification accuracy ranging from 88.6% to 94.5% under 5 dB to 20 dB SNR. This work bridges analog circuit design and machine learning, presenting a compelling solution for energy-efficient, compact, and accurate always-on KWS systems in mobile and IoT devices.

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